

A Local Move Principle for Constrained Expression in the Hilbert Substrate Framework

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Abstract

This note turns the current HSF “constrained expression” idea into an implementable local move principle. The goal is to replace ad hoc birth, pruning, and reabsorption rules with one objective functional, evaluated locally by counterfactual branch comparison. Subsystem birth, persistence, weakening, and reabsorption are treated as admissible local moves of the substrate. A move is preferred when it improves faithful constrained expression of the wavefunction more than it increases the burdens associated with finite bandwidth, no-refolding, no-signaling, and no-forgetting.

1 Guiding picture

The wavefunction is primary. Subsystems and links are not primitive ontology; they are temporary explicit support structures through which relational structure is carried and made operationally accessible.

Let R be a local region of the substrate containing:

- a small set of active site supports V_R ,
- a set of active links E_R ,
- the local reduced state ρ_R ,
- and any candidate dormant support patches adjacent to R .

At each update, the substrate considers locally admissible moves

$$M \in \mathcal{M}_R = \{M_0, M_+, M_-, M_{e+}, M_{e-}, M_{\text{swap}}\},$$

where:

- M_0 : no change,
- M_+ : add one explicit support patch / subsystem,
- M_- : demote one explicit support patch / subsystem,
- M_{e+} : strengthen or commit one interface / link,
- M_{e-} : weaken or decommit one interface / link,
- M_{swap} : redistribute support without changing total support count.

The preferred move is the one that best improves the *constrained expression functional* defined below.

2 Local constrained expression functional

For each local region R , define

$$\mathcal{F}_R = \mathcal{E}_R - \lambda_B \mathcal{C}_R^{(B)} - \lambda_R \mathcal{C}_R^{(R)} - \lambda_S \mathcal{C}_R^{(S)} - \lambda_F \mathcal{C}_R^{(F)}, \quad (1)$$

with nonnegative weights

$$\lambda_B, \lambda_R, \lambda_S, \lambda_F \geq 0.$$

Interpretation:

- $\mathcal{E}_R$: quality of expression of the relational structure actually present in R ,
- $\mathcal{C}_R^{(B)}$: finite-bandwidth burden,
- $\mathcal{C}_R^{(R)}$: no-refolding / commitment burden,
- $\mathcal{C}_R^{(S)}$: no-signaling / spread burden,
- $\mathcal{C}_R^{(F)}$: no-forgetting penalty.

A move M is evaluated by short-horizon branch comparison:

$$\Delta \mathcal{F}_R(M) = \mathcal{F}_R(\rho_R^{\text{moved}}(\tau)) - \mathcal{F}_R(\rho_R^{\text{base}}(\tau)), \quad (2)$$

where:

- $\rho_R^{\text{base}}(\tau)$ is the local state after evolving the unmodified branch for a short horizon τ ,
- $\rho_R^{\text{moved}}(\tau)$ is the local state after applying move M and evolving for the same horizon.

The simplest deterministic rule is:

$$M^* = \arg \max_{M \in \mathcal{M}_R} \Delta \mathcal{F}_R(M), \quad \text{accept } M^* \text{ if } \Delta \mathcal{F}_R(M^*) > 0. \quad (3)$$

A stochastic alternative is:

$$P(M \mid R) = \frac{\exp(\beta \Delta \mathcal{F}_R(M))}{\sum_{M' \in \mathcal{M}_R} \exp(\beta \Delta \mathcal{F}_R(M'))}, \quad (4)$$

with inverse-temperature parameter $\beta > 0$.

3 Concrete expression term $\mathcal{E}_R$

3.1 Overview

The expression term should reward explicit support and links when they improve the faithful representation of the relational structure actually present in the state.

We write

$$\mathcal{E}_R = w_{\text{MI}} E_{\text{MI}} + w_{\text{link}} E_{\text{link}} + w_{\text{shell}} E_{\text{shell}} + w_{\text{core}} E_{\text{core}}, \quad (5)$$

with nonnegative weights $w_{\text{MI}}, w_{\text{link}}, w_{\text{shell}}, w_{\text{core}}$.

3.2 Local mutual-information expression

Let $N_R(i)$ denote the graph neighborhood of site i restricted to R . Define

$$E_{\text{MI}} = \frac{1}{Z_{\text{MI}}} \sum_{(i,j) \in E_R} \omega_{ij}^{(\text{MI})} I(i:j), \quad (6)$$

where:

- $I(i:j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij})$,
- $S(\rho) = -\text{tr}(\rho \log \rho)$,
- $\omega_{ij}^{(\text{MI})} \geq 0$ are optional edge weights,
- $Z_{\text{MI}} = \sum_{(i,j) \in E_R} \omega_{ij}^{(\text{MI})}$ if the sum is nonzero, else 1.

This term says that a local support pattern expresses the wavefunction well when important active interfaces actually carry mutual information.

3.3 Link-expression term

Each active link $e = (i, j) \in E_R$ carries a link register state σ_e on

$$\mathcal{H}_{\text{link}} \cong V \otimes \bar{V}.$$

Let Φ_e be the local endpoint influence map for link e , defined operationally by probing endpoint i and reading response on the link, or vice versa. Let

$$s_1(\Phi_e) \geq s_2(\Phi_e) \geq \dots \geq 0$$

be its singular values.

Define three quantities:

$$r_e = \sum_a \mathbf{1}[s_a(\Phi_e) > \varepsilon_{\text{sv}}], \quad (7)$$

$$b_e = \frac{\sum_a s_a(\Phi_e)}{1 + \sum_a s_a(\Phi_e)}, \quad (8)$$

$$p_e = \text{tr}(\sigma_e^2). \quad (9)$$

Here r_e is the effective active-channel rank, $b_e \in [0, 1)$ is a bounded bandwidth-activity score, and $p_e \in (0, 1]$ is link purity.

Because HSF wants links to be *thin* rather than broad or slack-heavy, the expression reward should favor *active but selective* links. Define

$$E_{\text{link}} = \frac{1}{|E_R|} \sum_{e \in E_R} \left[\alpha_b b_e + \alpha_p p_e + \alpha_r \frac{1}{1 + r_e} \right]. \quad (10)$$

This rewards links that are:

- dynamically active (b_e not too small),
- coherent (p_e large),
- and thin/selective (r_e small).

3.4 Shell-support term

Let $C_R = (i_\star, j_\star)$ be the dominant local core pair in region R , and let $\text{Shell}(R)$ be the set of sites adjacent to at least one endpoint of C_R . Define

$$n_{\text{shell}} = |\text{Shell}(R)|, \quad (11)$$

$$n_\Delta = \#\{\text{triangles in } R \text{ touching } C_R\}, \quad (12)$$

$$\bar{d}_{\text{shell}} = \frac{1}{n_{\text{shell}}} \sum_{k \in \text{Shell}(R)} d_R(k), \quad (13)$$

where $d_R(k)$ is degree in the region.

Then define

$$E_{\text{shell}} = \alpha_{\text{shell}} \frac{n_{\text{shell}}}{n_{\text{shell}} + 1} + \alpha_\Delta \frac{n_\Delta}{n_\Delta + 1} + \alpha_d \frac{\bar{d}_{\text{shell}}}{\bar{d}_{\text{shell}} + 1}. \quad (14)$$

This term rewards support arrangements that form coherent mesoscale objects rather than isolated or dangling supports.

3.5 Core-expression term

For the dominant core pair $C_R = (i_\star, j_\star)$, define

$$E_{\text{core}} = \alpha_I I(i_\star : j_\star) + \alpha_\Gamma \Gamma(i_\star, j_\star) + \alpha_{b_\star} b_{(i_\star, j_\star)}, \quad (15)$$

where $\Gamma(i, j)$ is the chosen pair correlator strength (e.g. the norm of the vector of local generator-generator correlators).

This term rewards a strong dynamical core inside the local object.

4 Concrete bandwidth burden $\mathcal{C}_R^{(B)}$

Finite bandwidth should penalize *too many simultaneously active channels*, overstuffed links, and excessive explicit support.

Define

$$\mathcal{C}_R^{(B)} = u_r \left(\frac{1}{|E_R|} \sum_{e \in E_R} r_e \right) + u_n |V_R| + u_e |E_R| + u_s \rho_{\text{supp}}(R), \quad (16)$$

where

$$\rho_{\text{supp}}(R) = \frac{|V_R| + \eta |E_R|}{|R|_{\text{cap}}}$$

is local explicit-support density relative to the local capacity scale $|R|_{\text{cap}}$, and $\eta > 0$ weights edge occupancy.

Interpretation:

- more active channels r_e cost more,
- more explicit support patches cost more,
- more active interfaces cost more,
- overfilled local regions cost more.

5 Concrete no-refolding burden $\mathcal{C}_R^{(R)}$

No-refolding should penalize moves that rewrite or decommit already-working interfaces without preserving their relational role.

Let $E_R^{\text{comm}} \subseteq E_R$ be the set of committed interfaces in R . For each such interface e , define a commitment score

$$\kappa_e = \beta_b b_e + \beta_p p_e + \beta_{\text{age}} a_e + \beta_{\text{tri}} t_e, \quad (17)$$

where:

- a_e is edge age or persistence fraction,
- t_e is a local triangle/support score for that edge.

For a candidate move M , let $\delta_e(M) \in [0, 1]$ measure the fraction of commitment on edge e that is lost or incoherently reassigned under the move. Then define

$$\mathcal{C}_R^{(R)}(M) = \sum_{e \in E_R^{\text{comm}}} \kappa_e \delta_e(M). \quad (18)$$

In practice, $\delta_e(M)$ can be approximated by:

$$\delta_e(M) = 1 - \text{overlap}_e(M), \quad (19)$$

where $\text{overlap}_e(M)$ is a weighted overlap between the pre-move and post-move local shell/core role of edge e .

This term is high when a move destroys mature working interfaces rather than re-expressing them.

6 Concrete no-signaling / spread burden $\mathcal{C}_R^{(S)}$

No-signaling does not mean “no influence.” It means influence should remain bounded and not delocalize arbitrarily fast or broadly.

Let x_i be the current local metric coordinate of site i , obtained from a chosen local embedding (spectral or otherwise). Let c_R be the center of the dominant core. Define

$$\ell_i = |x_i - c_R|. \quad (20)$$

Then define

$$\mathcal{C}_R^{(S)} = v_\ell \left(\frac{1}{|V_R|} \sum_{i \in V_R} \ell_i \right) + v_\Phi \left(\frac{1}{|E_R|} \sum_{e \in E_R} \text{spread}(\Phi_e) \right), \quad (21)$$

where $\text{spread}(\Phi_e)$ is a measure of how broadly influence is distributed across the endpoint influence channels of the link.

A simple channel-spread measure is

$$\text{spread}(\Phi_e) = \frac{(\sum_a s_a(\Phi_e))^2}{\sum_a s_a(\Phi_e)^2}, \quad (22)$$

which is the effective rank of the influence map. This penalizes links whose influence is dispersed over too many channels.

7 Concrete no-forgetting penalty $\mathcal{C}_R^{(F)}$

No-forgetting should penalize loss of recoverable relational record. Demotion is allowed only if the same structure is adequately re-expressed elsewhere.

Define

$$\mathcal{C}_R^{(F)} = q_{\text{fid}} (1 - F_{\text{org}}) + q_{\text{MI}} [\Delta I_{\text{lost}}]_+ + q_{\text{loc}} \Delta_{\text{loc}}, \quad (23)$$

where:

- $F_{\text{org}} \in [0, 1]$ is organizer fidelity between baseline and moved local branches,
- $[\Delta I_{\text{lost}}]_+$ is the positive part of local MI loss,
- Δ_{loc} measures un-reexpressed loss of local shell/core support.

More explicitly, let

$$\Delta I_{\text{lost}} = \left(\sum_{(i,j) \in E_R^{\text{base}}} I^{\text{base}}(i : j) \right) - \left(\sum_{(i,j) \in E_R^{\text{moved}}} I^{\text{moved}}(i : j) \right). \quad (24)$$

Let

$$\Delta_{\text{loc}} = \left[\gamma_{\text{shell}} (n_{\text{shell}}^{\text{base}} - n_{\text{shell}}^{\text{moved}}) + \gamma_{\Delta} (n_{\Delta}^{\text{base}} - n_{\Delta}^{\text{moved}}) \right]_+. \quad (25)$$

This term is small when a move redistributes information and support while preserving the organizer-level state and local relational record.

8 Birth, demotion, and interface moves as one rule

8.1 Birth move M_+

Given a candidate parent interface $e = (i, j)$, define a birth move $M_+(e \rightarrow n)$ that creates a new support patch n and attaches it through one or more interfaces to the parent pair.

Birth is accepted when

$$\Delta \mathcal{F}_R(M_+(e \rightarrow n)) > 0. \quad (26)$$

Importantly, this does *not* require immediate burden relief. A birth may still be preferred if it produces enough improvement in constrained expression $\mathcal{E}_R$ to outweigh the burden increase.

8.2 Demotion move M_-

Given an active support patch n , define $M_-(n)$ as the move that demotes or reabsorbs the support patch while allowing link and shell structure to reorganize.

Demotion is accepted when

$$\Delta \mathcal{F}_R(M_-(n)) > 0. \quad (27)$$

Under this rule, reabsorption is *not* deletion. It is allowed only when the move relieves burden and preserves enough organizer-level relational structure that the net constrained expression improves.

8.3 Link commitment / decommitment moves

For an interface $e \in E_R$, link strengthening M_{e+} and weakening M_{e-} are accepted using the same rule:

$$\Delta\mathcal{F}_R(M_{e\pm}) > 0. \quad (28)$$

Thus links are not hard-coded as permanent or ephemeral. Their commitment is an emergent outcome of the same local move principle.

9 Minimal implementable version

The full functional above is the target. A minimal first implementation can use the following practical surrogate:

$$\mathcal{F}_R^{\min} = w_1 E_{\text{MI}} + w_2 E_{\text{link}} + w_3 E_{\text{shell}} + w_4 E_{\text{core}} - w_5 \mathcal{C}_R^{(B)} - w_6 \mathcal{C}_R^{(R)} - w_7 \mathcal{C}_R^{(S)} - w_8 \mathcal{C}_R^{(F)}, \quad (29)$$

with all $w_k \geq 0$.

Then, for each local move:

1. construct baseline and moved branches,
2. evolve both for a short horizon τ ,
3. evaluate $\mathcal{F}_R^{\min}$ on both,
4. compute $\Delta\mathcal{F}_R$,
5. accept the move with maximal positive $\Delta\mathcal{F}_R$.

10 Suggested coding choices

For a first implementation:

- choose R as the union of:
 - the candidate parent pair or candidate demotion site,
 - its immediate graph neighbors,
 - and the current dominant-core shell if overlapping.
- use a short horizon of 1–3 local evaluation windows,
- clip fidelity values to $[0, 1]$,
- normalize all terms to comparable scale before weighting,
- begin with deterministic best-positive-move dynamics,
- only later add softmax sampling.

11 Interpretive summary

This formalism replaces separate ad hoc rules with one principle:

Among locally admissible reorganizations, the substrate prefers the move that best improves the constrained expression of the wavefunction.

Birth, persistence, strengthening, weakening, and reabsorption are then not primitive laws. They are different local responses to the same objective.

12 Open calibration questions

This specification still leaves several things to be fixed empirically or theoretically:

1. the precise normalization of each term,
2. the best local definition of the influence map Φ_e ,
3. the best local metric or spread diagnostic for $\mathcal{C}_R^{(S)}$,
4. the exact surrogate for no-refolding overlap loss $\delta_e(M)$,
5. whether the dynamics should be deterministic or sampled.

But these are now calibration questions within one architecture, rather than separate hand-tuned rules for birth, pruning, and reabsorption.

Working bottom line

A move is admissible when it improves the local balance

faithful constrained expression – bandwidth burden – refolding burden – spread burden – forgetting penalty.

That is the proposed first-principles dynamical core for the next HSF mesoscape program.